IVF Lab Performance & Sperm Selection Analysis

Total_procedure

1197

Fertilization_Success_rate

59.57%

Average_selection_time

152.79

Usable_embryo_rate

23.98%

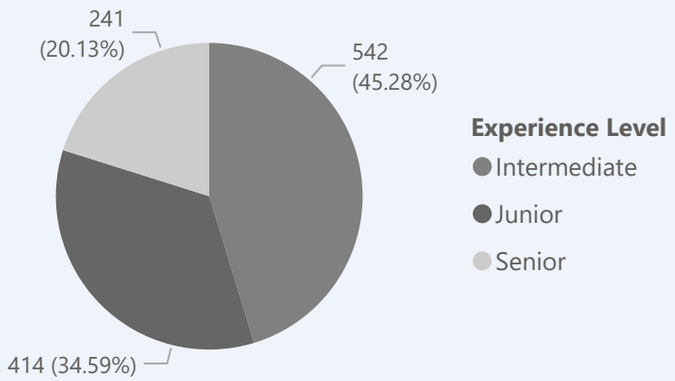
Embryologist_ID

All

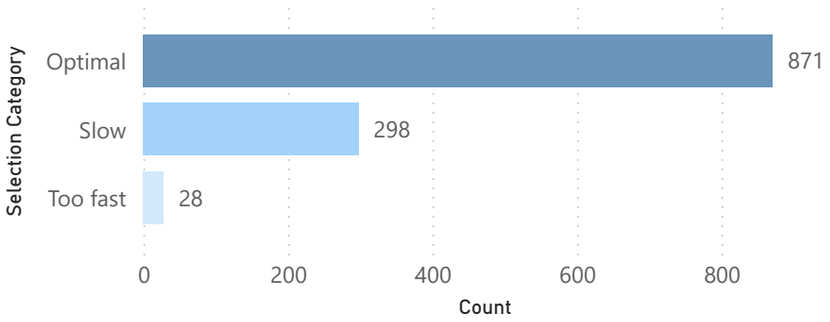
Experience_level

All

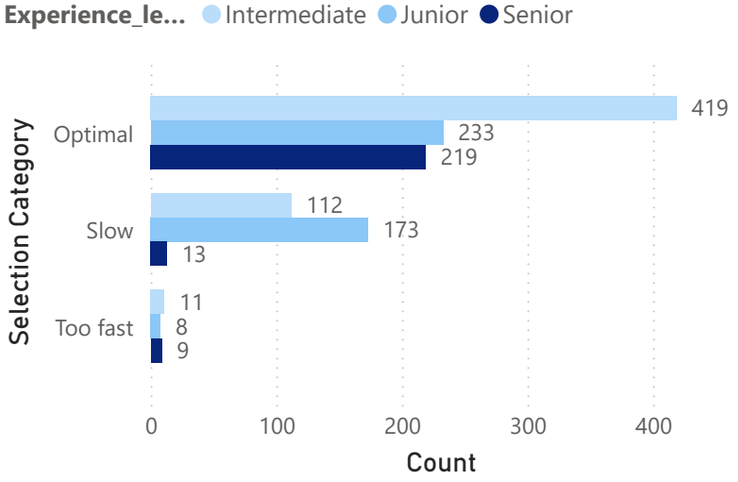
Contribution by Experience level



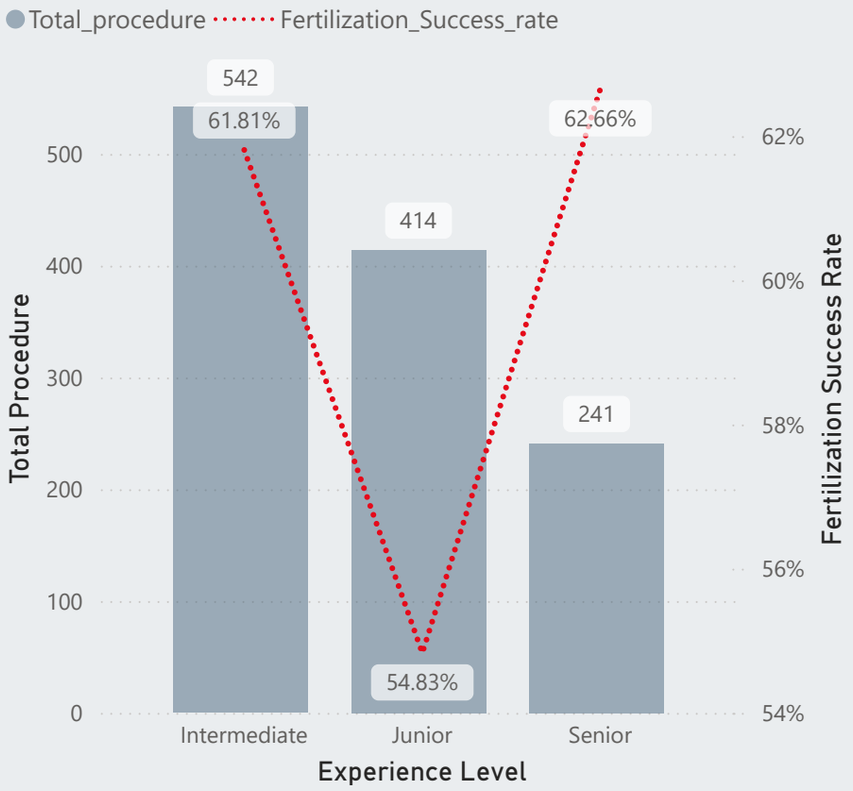
Selection time distribution



Selection time by Experience level



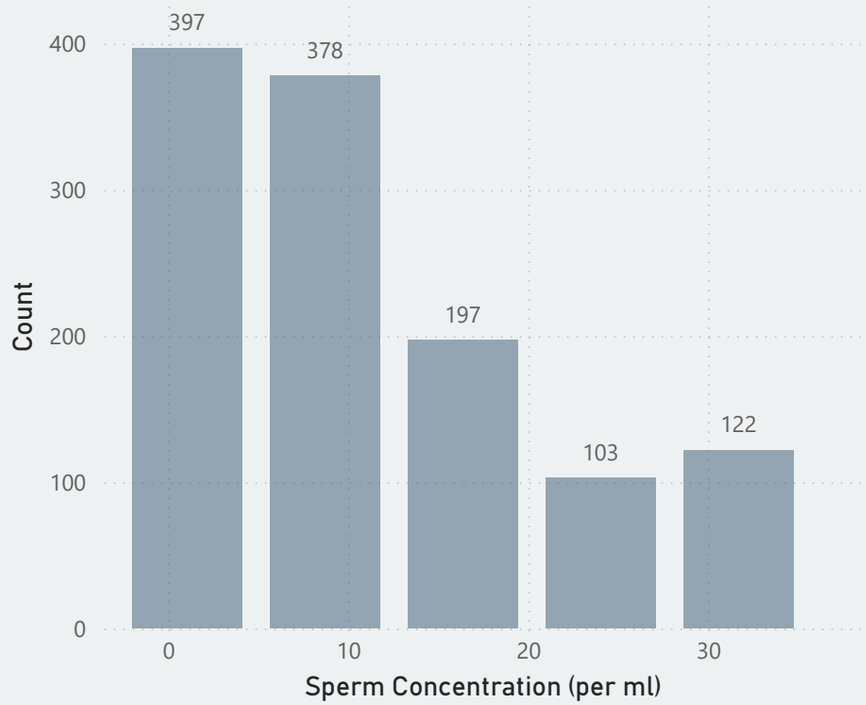
Total procedure & Success by Experience level



As the visuals represents, the IVF Lab currently has the most intermediate experience level embryologist with 45.28%, surprisingly performing well in optimal selection category. while lab has senior level embryologist @20.13%.

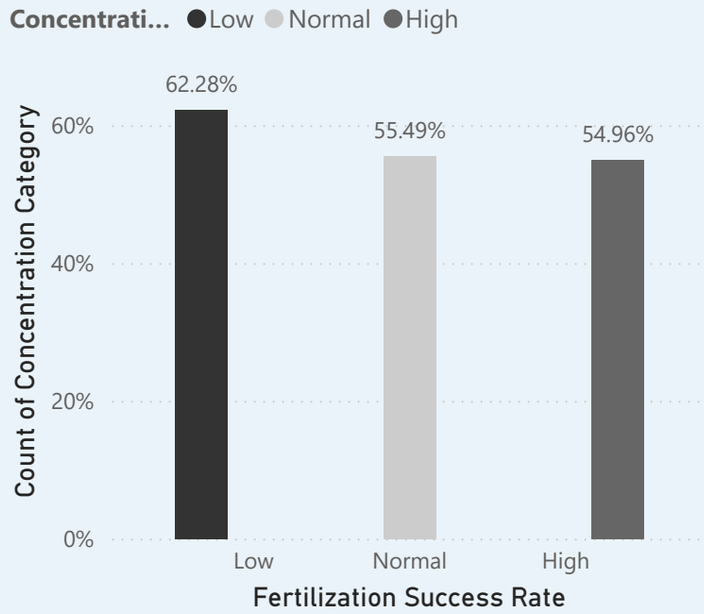
Total procedure & fertilization success rate by experience level indicates that senior level embryologist performed 241 procedure and has a success rate of 62.88%. The lab can assign senior level embryologist to complicated patients while intermediate and junior level patient can

Sperm concentration count

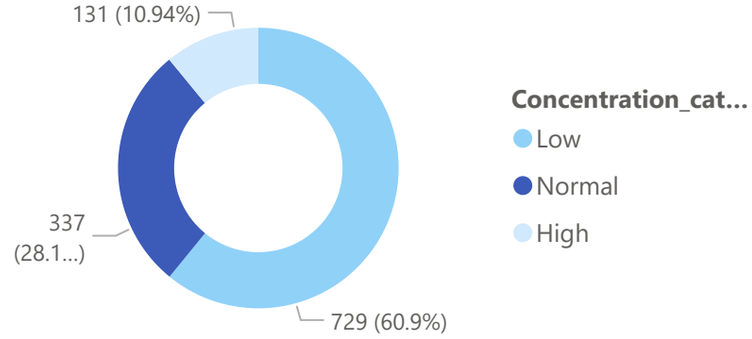


Here we are binning the values into 3 category 0-15 = low, 15-30 = normal & 30-40 = high.

Success rate by Concentration Category



Concentration category distribution



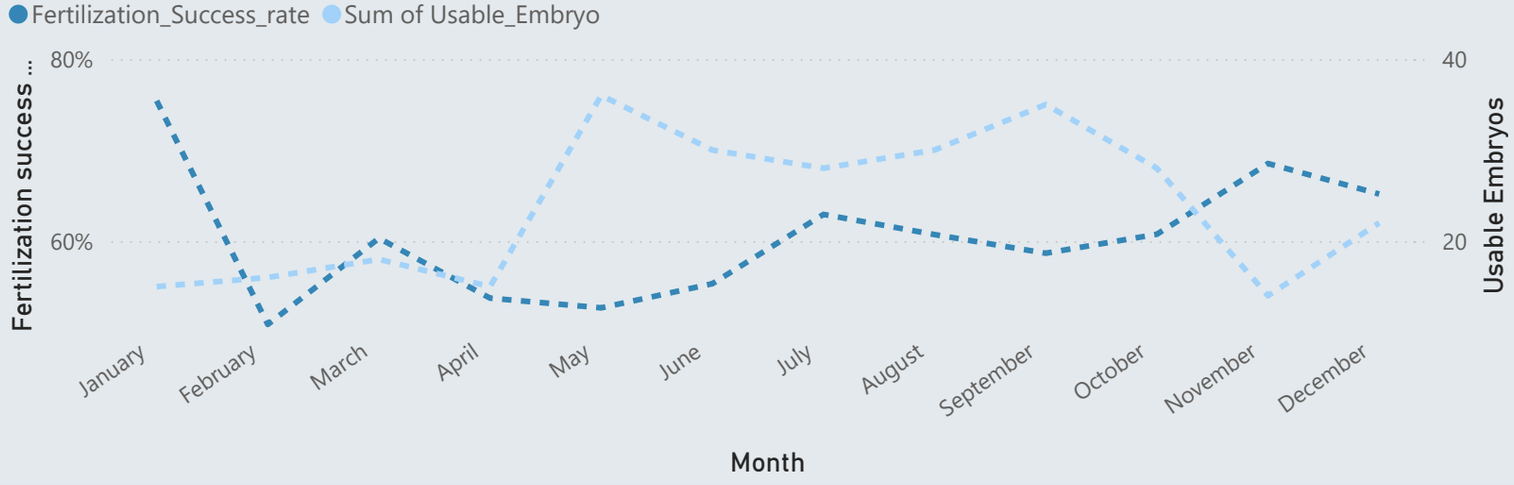
Looking at the value and visuals, LOW sperm concentration acquires 60.9%, and it has the highest success rate by 62.28%, while HIGH being the lowest success contributor with 54.96%.

Selection_Date

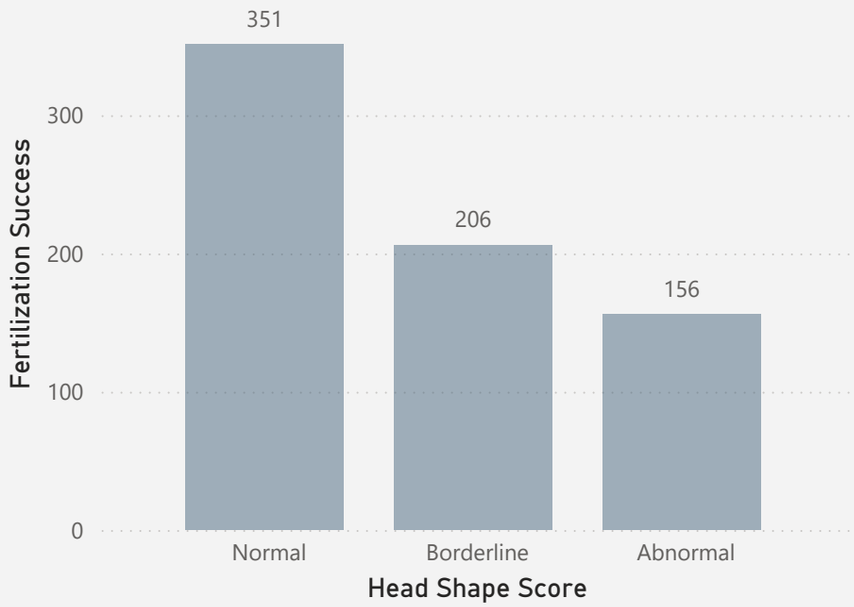
01-05-2024 17-12-2025

January is the most successful fertilization month with 75% and 15 usable embryos. Where as in May lab had acquired 36 usable embryo but fertilization success rate was only 52.67%. This indicates that even though we acquire good number of usable embryos, fertilization is still failing due to other factors.

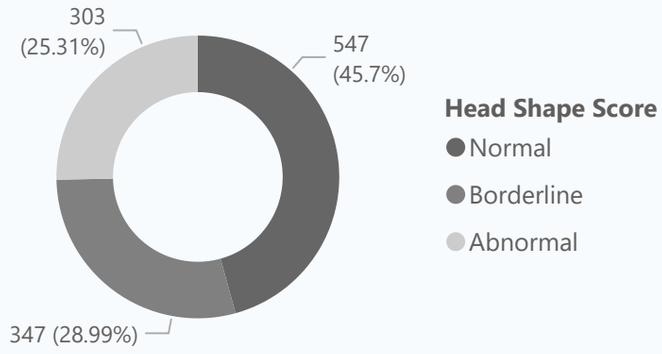
Fertilization success rate & Usable Embryo by Month



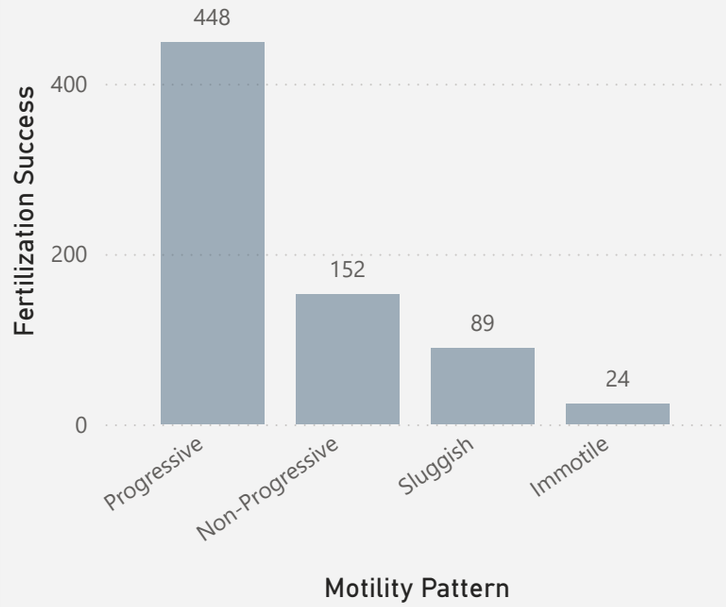
Fertilization success by Head shape score



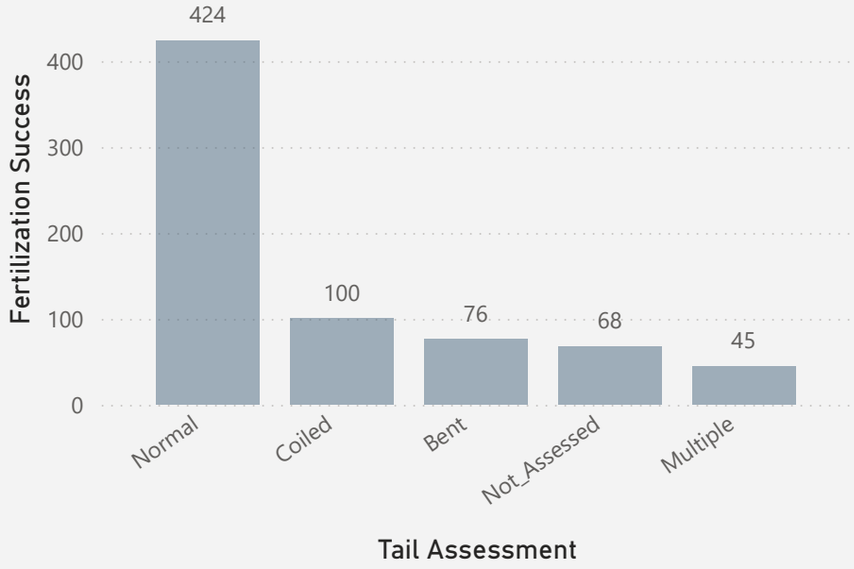
Head shape score distribution



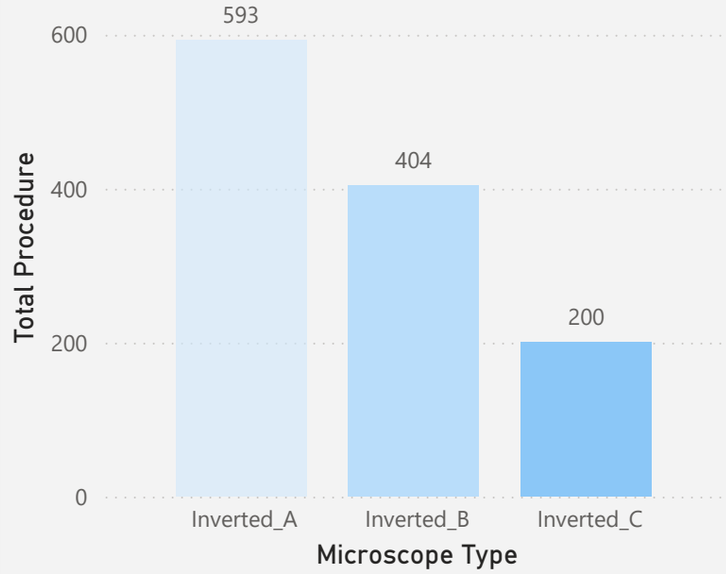
Fertilization success by Motility pattern



Fertilization success by Tail Assessment



Here, we can see that there is a high and positive correlation between progressive motility and head shape score with success rate of 64%. Hence, proving that higher clinical success is achieved by integrating advanced sperm quality metrics with standardized embryologist selection times.

Head_Shape_Score	Immotile	Non-Progressive	Progressive	Sl
Abnormal	35.29%	55.71%	54.39%	4
Borderline	53.33%	63.01%	61.03%	4
Normal	37.04%	61.47%	69.23%	5
Total	**40.68%**	**60.32%**	**63.19%**	**5**

Total Procedure by Microscope Type